$$p(r) = \frac{3\mu_0 I^2 (R^4 - r^4)}{16\pi^2 R^6}. \quad (22)$$

8. ,

$$w_\sigma = \frac{\sigma^2}{2E} = \frac{p^2(r)}{2E}, \quad (23)$$

$$W_\sigma = \int_0^R w_\sigma 2\pi r L dr = \frac{3\mu_0^2 I^4 L}{320 E \pi^3 R^2} = 2.39 \times 10^{-18} J. \quad (24)$$

9. .

$$\varepsilon = \frac{\sigma}{E} = \frac{p(r)}{E}, \quad (25)$$

ε .

$$\delta R_\sigma = \int_0^R \varepsilon dr = \frac{1}{E}\int_a^R p(r) dr = \frac{3\mu_0 I^2}{20\pi^2 E R} = 1.91 \times 10^{-12} m. \quad (26)$$

10. (15) (25)

$$\gamma = \frac{4\mu_0 \alpha\beta\kappa R^2}{5E(\alpha R + 4\kappa)} = 3.35 \times 10^{-10} K^{-1}. \quad (27)$$

채점 기준

№	평가 항목	배점	
1	(1)	0.25	
	(2)	0.25	
	(3)	0.25	**1.0**
	(3)	0.25	
2	(4)	0.25	
	(5)	0.25	
	(6)	0.25	**1,0**
	(6)	0.25	
3	(7)	0.25	
	(8)	0.25	
	(9)	0.25	
	(10)	0.25	
	(11)	0.25	**2.5**
	(12)	0.25	
	(13)	0.5	
	(14)	0.25	
	(14)	0.25	
4	(15)	0.25	**0.5**
	(15)	0.25	
5	(16)	0.25	**0.5**
	(17)	0.25	
6	(18)	0.5	
	(19)	0.25	**1.0**